Phases of Commercial Construction

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Course Objectives

After completing this course, learners will be able to:

- Understand how a commercial building goes together
- Define the phases of construction
- Identify who's on the job at each phase, and what they're there to do

- Sections or Chapter Slides
- Activity Slides
- Knowledge Check Slides
- Video

Slide #	Slide title	Narration	Onscreen slide content	Graphics/media	Notes
N/A	**Resources**	N/A			
1.0	**Introduction**				
1.1	**Logo**	N/A	N/A	Animated Company Logo	
1.2	**Introduction**	From breaking ground to the grand opening, a commercial construction jobsite is a busy place. Professionals from nearly every trade in the industry are there at one time or another to	This course contains audio narration.	Time lapse showing construction site with many people moving to and fro https://www.shutterstock.com/video/clip-3629798-stock-footage-building-construction-site-with-workers-hard-at-work-	

		complete their portion of the build. With all that activity, have you ever wondered how everyone knows when to show up, where to go, and what to do?		in-a-city-downtown-timelapse-x.html?src=search/OBte47KGwH3uJcTG0z9Aog:1:17	
1.3	**Phases of Commercial Construction**	The answer is phases of construction. Most large-scale projects are broken down into a series of smaller, sequential phases, based on overall project workflow. It starts with **planning** and approvals, followed by **site prep** and excavation. Once the site has been readied, a **foundation** or basement is poured to support a **structural** frame. Next, **mechanicals** go in, then the **exterior** shell goes on, so that final **interior** finish work can begin. Once everything is built and **landscaped**, a final **punchout** inspection ensures everything is complete. This is a logical sequence. Each new phase begins upon completion of the previous. It wouldn't make sense to pour a foundation before the project's been approved, or hang drywall	**Phases of Commercial Construction** Planning Phase Site Prep Phase Foundation Phase Structure Phase MEP Phase Exterior Finish/Dry-in Phase Interior Finish Phase Landscaping Phase Punchout Phase	Workflow diagram showing all of the phases in sequence, with images symbolic of each stage.	**Designer:** This is the title slide.

		before the mechanicals are roughed-in.			
1.4	**Objectives**	Today, you'll learn how new commercial buildings come together by following a phased construction process. We'll define each phase, so that by the end of this course, you'll have a better idea of who's on the jobsite, when, and what they do at each stage of the process. Click 'NEXT' to go on.	**Today's Objectives** • Understand how a building comes together • Define the phases of construction • Identify who's on the job at each phase		
2.0	**Planning Phase**				
2.1	**Phase 1: Planning**	Phase One: Planning. During the planning phase, the project owner works with the local zoning commission to select a location for construction, while architects and engineers work together to conceptualize the structural design. Once the site and the structural design plans are approved, permits are issued and project bidding begins!	**Phase One: Planning** ✓ Building site selected ✓ Structural design conceptualized ✓ Approvals and permits issued ✓ Bidding begins	Image: Construction site before breaking ground or of collaboration around site planning, structural design --OR-- Video: https://www.shutterstock.com/video/clip-2497598-stock-footage-presentation-of-new-modern-building-design-project.html?src=recommended/3421214:7	
2.2	**Project Personnel**	As the project gets underway, more people get involved. Clearly defined roles and	**Project Personnel** Project Owner	Aerial shot of construction site with people. Hotspots on people. Mouse over each person's silhouette and then get a badge	**Hotspot Activity**

		responsibilities make it easier to identify who does what, and who's in charge. Click on the markers to learn about the different **roles** on the jobsite now.	General Contractor Safety Superintendent Trade Superintendent Trade Foreman Trade Laborers Click on the markers to learn more.	title. Badges with images, job titles, and responsibilities pop up.	
2.2A	**Wait**	N/A	**Wait.** There's more to learn here. [GO BACK]		
2.2B	**Project Owner**	Project owners are responsible for overseeing the entire project. The project originates when the owner selects a site, has it surveyed, and calls for design plans to get drawn up. They hire the engineer and general contractor, and provide all of the financing. So, it makes sense that they also approve the budget, design, schedule, and materials. They're also responsible for communicating critical information about the project with key stakeholders, including regulatory agencies and real estate professionals when it's time to sell or lease the finished structure.	**Role: Project Owner** **Responsibilities:** • Selects the site • Calls for site surveys • Commissions design plans • Hires architect and general contractor • Provides financing • Approves budget, design, schedule, and materials • Communicates with project stakeholders, including regulatory agencies and real estate professionals	Badge w/photo, title, and responsibilities	

2.2C	**General Contractors**	General Contractors, who you'll also sometimes hear called project managers, are responsible for managing the project and overseeing the construction site on a day-to-day basis. They manage project documentation, including blueprints, and set the master schedule. They are responsible for hiring subcontractors and procuring the necessary materials and equipment to get the project completed on time, and on budget. They work closely with project owners to get approvals and secure payment for completed work.	**Role: General Contractor** **Responsibilities:** • Day-to-day project management and oversight • Manages blueprints • Sets master schedule • Hires subcontractors • Procures materials & equipment • Keeps project on time, and on budget • Works with project owner to get approvals and secure payment for completed work	Badge w/photo, title, and responsibilities	
2.2D	**Safety Superintendent**	The Safety Superintendent is responsible for ensuring the safety of everyone on the jobsite. Their job is to enforce jobsite safety requirements and provide necessary training. Because a construction site is a dangerous place, and the consequences for non-compliance are severe, Safety Superintendents function like gatekeepers to the jobsite. If you don't have the proper protective equipment, permission, or basic safety smarts, they won't let you get far.	**Role: Safety Superintendent** **Responsibilities:** • Ensures everyone's safety • Enforces safety requirements • Provides safety training • Serves as a gatekeeper to the project site	Badge w/photo, title, and responsibilities	

2.2E	**Trade Superintendent**	On large jobsites and projects, Trade Superintendents function as intermediaries between the general contractor and trade laborers. They act as supervisor for their specific trade, delegating daily tasks to crew, and ensuring that assigned tasks get done on time and on budget. They report to the general contractor to provide updates on their progress and communicate any problems or concerns that could halt or delay the project.	**Role: Trade Superintendent** **Responsibilities:** • Serves as a liaison between general contractor and trade laborers • Performs supervisory duties within their specific trade • Delegates daily tasks • Ensures work is completed on time and on budget • Reports to general contractor on progress and problems that could halt the project	Badge w/photo, title, and responsibilities	
2.2F	**Trade Foremen**	Trade Foremen have specialized knowledge of their trade. Their many years of experience qualify them as a source of expertise for members of their crew. As a result, they have a supervisory role on a jobsite, and help with training those with less experience on the safe and proper use of equipment. They may also be responsible for setting the schedule for their crew and arranging for the delivery of materials. Usually, there will be just one foreman onsite for each trade, but there could potentially be multiple	**Role: Trade Foreman** **Responsibilities:** • Shares specialized knowledge and industry experience • Provides training • Assists with scheduling • Arranges for delivery of materials	Badge w/photo, title, and responsibilities	

		foremen overseeing groups within that trade, such as a rough crew or trim crew.			
2.2G	**Trade Laborers**	Trade Laborers perform tasks specific to their trade. They regularly use hand and power tools, and heavy equipment like cement mixers and hoists. They don't have a lot of decision making power or authority, but they are not entirely without influence. For example, their preferences for using a specific brand of tools or building materials can inform engineering specs and purchasing decisions.	**Role: Trade Laborers** **Responsibilities:** • Work on tasks specific to their trade • Performs physical labor • Works with tools and equipment • Communicates preferences that can influence project decisions	Badge w/photo, title, and responsibilities	
2.3	**[Placeholder] Knowledge Check**	Let's pause now to see what you've learned so far. Which of these individuals is responsible for selecting the construction site and approving the project budget? Mark the correct response, then click 'SUBMIT'.	**Knowledge Check** Which of these individuals is responsible for making sure everyone uses the proper protective equipment while on the jobsite? Project Owner General Contractor Safety Superintendent Trade Foreman		
2.3A	**KC1 Feedback Correct**	That's correct!	**Correct!** ☒ Project Owner		

		The Safety Superintendent is responsible for making sure everyone knows and follows safety protocols while on the jobsite.	☒ General Contractor ☑ Safety Superintendent ☒ Trade Foreman		
2.3B	**KC1 Feedback Incorrect**	That is incorrect. The Safety Superintendent is responsible for making sure everyone knows and follows safety protocols while on the jobsite.	**Incorrect.** ☒ Project Owner ☒ General Contractor ☑ Safety Superintendent ☒ Trade Foreman		
3.0	**Early Phases of Construction**		**Early Phases of Construction**		
3.1	**Phase 2: Site Prep**	Phase Two: Site Prep & Excavation. During the site prep phase, a survey company marks the site using existing landmarks and tags. Then, an excavation company brings in heavy equipment to prep the land for the foundation. In the case of very large or high-rise buildings, the site must be excavated down to the bedrock in preparation for caissons to be poured.	**Phase Two: Site Prep** ✓ Survey site ✓ Excavate land ✓ Trench-in electrical ✓ Install underground pipes ✓ Dig in water and sewer lines (or drill wells and prep for septic system)	Image of land excavation, land survey --OR-- https://www.shutterstock.com/video/clip-14194811-stock-footage-aerial-flight-over-earth-moving-machines-and-construction-site-construction-crews-prepare-a-site.html?src=search/xaiFaSGyQgYelC6xwi9RlQ:1:7 Icons representing the following trades: ➨ Survey / Measuring ➨ Excavation ➨ Utilities (Electrical, Plumbing, Sewer, Water, Gas)	

		Some underground utility work also happens during the site prep phase: • Electrical power is trenched into the lot. • Underground pipes are brought in to carry natural gas. • And, water and sewer lines are also dug in at this time, including prep work drilling wells and installing septic systems.			
3.3	**Phase 3: Foundation**	Phase Three: Foundation. Next, a concrete contractor begins to form up the foundation slab or the caissons that will support the weight of the building. But concrete work is not the only thing happening at this phase. This is also the time when initial utilities get "roughed in." Electrical contractors are on site to run main power lines that will support the future structure. And plumbing contractors are there to dig and run underground drainage, and rough-in waste and water supplies.	**Phase Three: Foundation** ✓ Form foundation or concrete footings ✓ Run power under the concrete slab ✓ Rough-in waste and water supplies ✓ Dig underground drainage, install sump pumps ✓ Form foundation walls ✓ Backfill dirt around the foundation ✓ Rough grade the lot	Image of foundation being poured, OR: https://www.shutterstock.com/video/clip-14194811-stock-footage-aerial-flight-over-earth-moving-machines-and-construction-site-construction-crews-prepare-a-site.html?src=search/xaiFaSGyQgYelC6xwi9RlQ:1:7 Image of rough-in, OR: https://www.shutterstock.com/video/clip-4521566-stock-footage-laying-pipe.html?src=search/yCGsk_oB7Fx_3oq0hm8E4w:1:4 Icons representing the following trades: ➨ Concrete / Masonry ➨ Utilities – Electrical, Plumbing, Sewer, Water, Gas	

		Once the foundation walls are set, the excavator returns to "back fill" dirt around the foundation, and rough grade the lot.		➨ Excavator	
3.4	**Phase 4: Structure**	Phase Four: Structure. Who shows up to the job site next depends on the type of structure being built. Framing materials are also driven by the type of structure. • If it’s a steel frame structure, or if steel is being used for reinforcement, iron workers will set to work. • If the building design requires wood framing, carpenters will be there. • If the structure is to be made of concrete, then concrete contractors will no doubt have a heavy presence. Depending on the design, crews from each of these trades could be working at the same time!	**Phase Four: Structure** Steel Frame Wood Frame Concrete Frame Multi-story Pour	Image of each type of structural frame (steel frame, wood frame, concrete)   Add similar picture for a tilt up building Add a similar picture for a multi-story pour Icons representing the following trades: Iron worker Carpenter	

				Concrete	
3.5	**Commercial Construction Methods**	The structure phase is where you'll see the most deviation between construction methods. Let's take a closer look at how today's most common commercial structures are assembled. Click on each construction method to learn more about it.	**Commercial Construction Methods** Tilt-up or Precast Construction Steel Frame Construction Multi-Story (Slip or Jump Frame) Construction Click to learn more.	Thumbnail for each type of construction method.	
3.5A	**Wait**	N/A	**Wait.** There's more to learn here. [GO BACK]		
3.5B	**Tilt-Up**	Tilt-up, or pre-cast, construction, is a popular concrete construction choice for warehouses and large "box" retailers, like Home Depot and Lowe's. Building components including walls, columns, and supports are made of formed concrete. These may be poured and cured on site, or cast offsite and trucked in. Cured components are lifted, or "tilted-up", into place with a crane and braced into position by	**Tilt-Up Construction** Building components are made of formed concrete. Cured components are lifted into place and braced into position.	Images of commercial building constructed / being constructed using tilt-up method OR https://www.shutterstock.com/video/clip-7998946-stock-footage-panel-house-construction-crane.html?src=search/A5SzgDbSF4vZ-oKpZfb0vw:1:12	

		concrete workers. Steel plates cast into the concrete are then welded together by iron workers to hold the components in place. Once the components start going up, the shell of the structure quickly takes shape! It's an economical and efficient choice for large structures, best suited to basic, rectangular designs.	Steel plates are welded together to hold components in place. An economical and efficient choice for large structures, best suited to basic, rectangular designs.		
3.5C	**Steel**	Steel frames formed the bones of the first skyscrapers, and many that came after. Today's steel constructed buildings are typically just one to four stories tall. It's a popular choice for hotels, storage facilities, medical centers, warehouses, sports complexes, and *small* retail and office buildings. In this type of construction, ironworkers weld prefabricated steel beams together to form the structural frame and carry the weight of the floors, ceilings, and walls that will come later. Because it has properties to span farther than wood, steel offers incredible versatility to accommodate modern architectural designs.	**Steel Frame Construction** Prefabricated steel beams are welded together to form the structural frame. Floors, ceilings, and walls come later. Offers incredible versatility to accommodate modern architectural designs.	Images of commercial building constructed / being constructed using steel frame method OR https://www.shutterstock.com/video/clip-11772608-stock-footage-video-shot-of-a-construction-site-steel-framing-building.html?src=/Xh-kLlcBaDFxJwm74-i95g:4:16	

3.5D	**Multistory**	Whereas their historic counterparts were forged with steel frames, today's high-rise buildings are built using the multi-story pour construction method. With this type of project, multiple phases of construction happen simultaneously. Here's how it works. Concrete workers pour a steel-reinforced concrete core using a removable "slip frame" or "jump frame" form. As each section of the core cures, the form is removed, lifted, and the process is repeated on a higher floor. As the core climbs higher into the sky, the project continues to move forward through the remaining phases of construction on the lower floors. With work happening in multiple phases simultaneously, it's an efficient method for keeping projects of such magnitude moving continuously.	**Multi-story Construction** A steel-reinforced concrete core is poured using a removable frame. As each section of the core cures, the process is repeated on a higher floor. The project moves on to the remaining phases of construction on lower floors. An efficient method for keeping projects of such magnitude moving continuously.	Images of commercial building constructed / being constructed using multi-story pour method OR https://www.shutterstock.com/video/clip-5456564-stock-footage-cranes-at-a-skyscraper-construction-p-time-lapse-video.html?src=search/bUcFxEn2QZ97mOKGVqA5Zw:1:1	
3.6	**Knowledge Check 2**	Let's pause again for another knowledge check.	Concrete workers are the only professionals on site during the foundation stage. True		

		Read the statement and mark whether it's true or false. Click 'SUBMIT' for feedback.	False		
		That's right! Concrete workers are ***not*** the only professionals on site during the foundation stage. The mechanical trades are also on site during the foundation stage to install underground pipe for things like electric, gas, water and waste.	**That's right!** Concrete workers are ***not*** the only professionals on site during the foundation stage. The mechanical trades are also on site during the foundation stage to install underground pipe for things like electric, gas, water and waste. In fact, multiple trades are present on site during ***every*** phase of construction.		
		That is incorrect. Concrete workers are ***not*** the only professionals on site during the foundation stage. The mechanical trades are also on site during the foundation stage to install underground pipe for things like electric, gas, water and waste.	**That is incorrect.** Concrete workers are ***not*** the only professionals on site during the foundation stage. The mechanical trades are also on site during the foundation stage to install underground pipe for things like electric, gas, water and waste.		

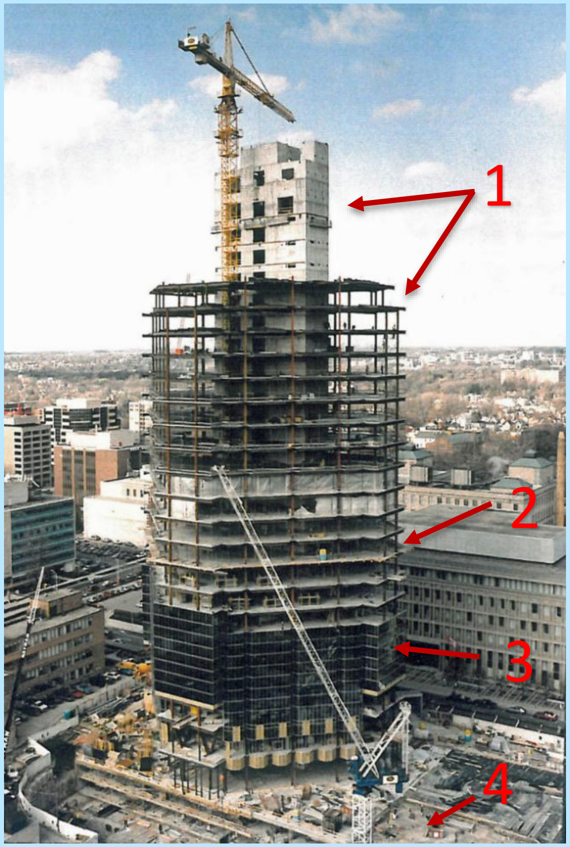
			In fact, multiple trades are present on site during ***every*** phase of construction.		
4.0	**Late Phases of Construction**				
4.1	**Phase 5: MEP**	Phase Five: MEP After the structural frame is in place, the MEP—or Mechanical, Electrical, and Plumbing—trades come in and start the "mechanicals" phase. If the structural frame is the bones of the building, these are its guts. Ductwork and ventilation takes the most space, so the HVAC contractor is usually first to get started. Next, the electrical contractor runs conduit, BX, and Romex cable throughout the building, which will later be used to power everything from electrical outlets to emergency warning systems. Meanwhile, plumbers install drainage, and run hot and cold water supplies to bathrooms, kitchens, and mechanical rooms. Where fire protection is	**Phase Five: MEP** ✓ Install HVAC ductwork ✓ Run electrical conduit, BX, and Romex cable ✓ Run power for emergency systems ✓ Run water supplies to bathrooms, kitchens, and mechanical rooms ✓ Install drainage ✓ Run PVC pipe and install sprinklers for fire protection	Images of MEP professionals working inside of a partially constructed building OR https://www.shutterstock.com/video/clip-33963652-stock-footage-a-number-of-five-white-vent-pipes-go-from-floor-to-ceiling-and-converge-into-a-common-channel.html?src=search/WIhlQQc50Y37LhdcJ-uQeQ:1:11 Trade icons for: HVAC Electrical Plumbing	

		required, pipefitters run pipe and install sprinklers at this stage.			
4.2	**Phase 6: Dry-In**	Phase Six: Exterior Finish/Dry-In While the mechanical trades are working on the inside, the exterior "dry-in" and finish work begins on the outside. During this phase, exterior facades—which can be glass, metal, or stone panels—are installed, as are exterior windows and doors. Depending on the type of exterior finish, any number of specialty contractors could arrive to begin applying exterior cladding. For example, a siding contractor would apply vinyl, wood, or aluminum cladding; a glazer installs glass curtain walls; but a masonry worker would do brick or stone finish work.	**Phase Six: Exterior Finish/Dry-In** ✓ Install exterior façade paneling ✓ Install exterior windows and doors ✓ Install exterior trim and cladding	Image of building with moisture barrier wrap showing, some siding installed. OR Trade icons: Carpenter Specialty Contractor > Roofing, Siding, Stonemasons	
5.0	**Finish Work**				
5.1	**Phase 7: Finish**	Phase Seven: Interior Finish After the exterior is "dried-in", interior finish work can begin.	**Phase Seven: Interior Finish** Specialty Finish Work MEP Finish Work	Possible slider or click to reveal activity (with clickable objects locked so that the learner must proceed in sequence)	**Activity Slider or Click to Reveal**

		Drag the slider to learn more about the work that gets done at this stage to push the project to completion.	Exterior Finish Work Drag the slider to learn more.		
5.1A	**Wait**	N/A	**Wait.** There's more to learn here. [GO BACK]		
5.1B	**Specialty Finish Work**	Many specialty contractors are on site to do finish work. An insulation contractor attaches insulation in walls and ceilings, to mitigate thermal loss. Once insulation is in place, ceilings are installed. The finish carpenter's work can include building stairs, framing walls, and installing interior windows, doors, and cabinets. With these things in place, the drywall contractor can begin hanging drywall, and taping and mudding seams in preparation for the painter. In addition to painting the interior walls, the painting contractor is also responsible for painting ceilings and trim, and in some cases, exterior painting as well.	**Specialty Finish Work** ✓ Install insulation ✓ Build stairs, frame walls, install interior doors and windows ✓ Hang drywall, tape and mud seams ✓ Paint walls and trim (Interior and exterior) ✓ Install floors and any backsplashes	Image of partially completed interior (walls and insulation) Trade Icons: Specialty Contractors = • Insulation • Drywall • Doors/Windows • Painting • Flooring https://www.shutterstock.com/video/clip-29240698-stock-footage-ethnic-worker-painting-walls-in-a-hall-with-roller-camera-on-glidecam-moving-around.html?src=search/MbE9spMAnvFkSowvstFwjg:1:33 Do we want any video showing what else gets done at this stage?	

		When the paint dries, flooring contractors set to work installing flooring. Some flooring contractors may have stone and tile installers on staff. Other times, specialty tile and stone contractors install the finish for bathroom walls and backsplash.			
5.1C	**MEP Finish Work**	Each of the mechanical trades have a trim phase as well. Plumbers install sinks and toilets, and hook up water systems. Electricians install specialty lighting, outlets, and switches, and connect any specialty appliances that are not directly plugged in. The HVAC contractor installs vent covers and thermostats at this time, as well as any special dehumidifiers, humidifiers, and air purification systems.	**MEP Finish Work** ✓ Install sinks and toilets, hook up water systems ✓ Install lighting, outlets, switches, and appliances ✓ Install HVAC vents, thermostats, dehumidifiers and air purification systems	Image of partially completed interior (wiring, sink) Trade icons for: Plumbers Electrical HVAC Need video for this?	
5.1D	**Exterior Landscape & Hardscape**	As the project nears completion, exterior landscaping and hardscaping can begin. This can include a final grade made by the original excavator. Concrete contractors may form poured concrete sidewalks and paved driveways, whereas masonry contractors would install pavers and retaining walls.	**Exterior Finish** ✓ Final grading ✓ Pave sidewalks and driveways ✓ Landscaping	Image of building exterior, almost finished, landscaping in progress Trade icons: Excavation Masonry / Stone / Concrete Landscaper Need video for this?	

		Finally, a landscaper arrives to plant trees, bushes, and grass.			
5.2	**Punchout**	All aspects of construction must pass inspection at each phase. Additionally, at the end of the project, everyone must complete a final punch out to make sure that items that have been flagged on previous inspections are now fixed. A project is not considered complete until the owner, the superintendent, and any regulatory agencies have signed off on it to ensure that all work is done perfectly, and with a clean finish.	**Final Phase: Punchout** ✓ All previously flagged items are fixed ✓ Project complete!	Image of completed structure, final inspection in progress	

5.3	**[Placeholder] Knowledge Check**	Take a look at this image of a high rise building under construction, where multiple phases are happening simultaneous. Identify each phase of construction by dragging the labels into the space provided. Click 'SUBMIT' to see if you're correct.	Drag the name of each phase of construction into the space provided to label the image. Then click the SUBMIT button. [1] Structure [2] MEP [3] Exterior Finish "Dry-In" [4] Foundation		**Drag & Drop:** Drop targets will be boxes with lines pointing to the different parts of the building / different phases. Learner will drag labels (textboxes) to the drop target to label the image. **Designer**: Drag items return to starting location if dragged to incorrect drop target.
5.3A	**Knowledge Check Feedback**	Excellent!	**Excellent!**		
6.0	**Conclusion**				

6.1	**Key Takeaways**	Today, you've learned how a commercial construction project goes from planning to punchout, and who's involved at each step along the way. A phased approach keeps a project moving forward and keeps everyone on the same page. The kind of work that happens at each phase determines who will be on the jobsite at any given time. Some individuals, like the Safety Supervisor, are there at every step of the way. But, there are many individuals who come and go only when they are needed to do work specific to their trade. As you've seen today, you can expect to encounter a range of professionals from across many trades as the project moves from excavation to rough-in to MEP and finish work.	**Key Takeaways** A phased approach keeps a commercial construction project moving forward and keeps everyone on the same page. The kind of work that happens at each phase determines who will be on the jobsite at any given time. You can expect to encounter a range of professionals from across many trades as the project moves through each of the phases.	Revisit time lapse video? Bring back all the icons?	
6.2	**Thank You**	Thank you for your time and participation today. This concludes the training. You may now close this window and launch the quiz.	**Thank You!**	Similar treatment as title slide.	

LMS Assessment

Q #	Type	Question	Responses	Feedback	Notes
1	**Multiple Choice**	Which of these does NOT take place during the planning phase?	Building site selected Structural design conceptualized Approvals and permits issued Land surveyed and excavated	Land surveys and excavation happen during the excavation phase, which begins once the planning phase is complete.	
2	**True / False**	Concrete workers are the only professionals on site during the foundation stage.	True False	Multiple trades are present on site during ***every*** phase of construction.	
3	**Multiple Choice**	Which trade would have the heaviest presence on the jobsite during the structural phase of a steel frame construction project?	Iron Workers Concrete Workers Carpenter Contractors Stone Masons	For steel frame construction projects, ironworkers weld pre-fabricated pieces of steel together during the structural phase.	
4	**Multiple Choice**	Which construction material is best suited for accommodating modern architectural designs?	Wood Steel Concrete	Steel offers incredible versatility to accommodate modern architectural designs.	
5	**Multiple Choice**	Multi-story pour construction method is used for:	Residential Homes Warehouses Sports Complexes High-rise Buildings	The multi-story pour construction method is an efficient approach for keeping high-rise projects continuously moving forward.	

6	**Multiple Choice**	Which of these would be used to support a pre-cast concrete commercial building?	Slab Foundation Caisson Footings	Pre-cast commercial buildings are constructed on top of a concrete slab foundation. Caisson footings are used to support multi-story high-rise structures.	
7	**Multiple Choice**	MEP stands for:	Mechanical, Electrical, and Plumbing Mapping, Engineering, and Planning Mitigation, Expansion, and Preparation Modifications, Extras, and Punchout	The MEP phase is when the building "mechanicals" go in, including Mechanical, Electrical, and Plumbing.	
8	**Multiple Choice**	When does a building get weatherproofed?	During the Structural Phase During the Exterior Finish/Dry-in Phase During the Landscaping & Hardscaping Phase During the Punchout Phase	Barrier wrap is taped and sealed up to protect the building from moisture during the Exterior Finish/Dry-in Phase.	
9	**Multiple Choice**	Which of these trades would be present during the Interior Finish phase?	Specialty Contractors MEP Contractors Landscapers All of the Above	Contractors from many trades are on site during the Interior Finish Phase, including all of those listed here!	
10	**Multiple Choice**	What happens during the punchout phase?	Excavators dig down to the bedrock in order to pour caisson footings. MEP contractors rough-in the mechanicals, electrical, and plumbing. The project owner, superintendent, and regulatory agencies sign off on the project. The project owner works with real estate professionals to sell the building.	During the Punchout Phase, the project owner, superintendent, and regulatory agencies sign off on the project.	

--	**Multiple Choice**	What phase of construction is shown here? [IMAGE]	Planning Phase Site Prep Phase Foundation Phase Structure Phase MEP Phase Exterior Finish/Dry-in Phase Interior Finish Phase Landscaping Phase Punchout Phase	This is an example of work done during the ________ phase.	